

Practice set 4

Merit and Excellence problem solving

Write in the spaces

Answers · official assessment schedule extracts on the last pages

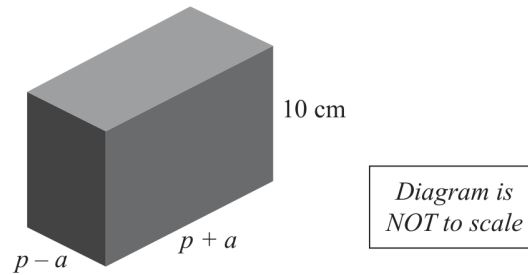
- (d) Consider the following two curves:

$$x^2 = y^2 + 1 \text{ and } y = (x - 1)(x + 1) - 2$$

Find the co-ordinates of each intersection point of the two curves.

- (d) Junyang realises that the **surface area** of the cuboid will not be the same as the surface area of the cube unless he also changes the height. He decides to make the height of the cuboid 10 cm.

He wants to find out which value of p would result in the cube having the same surface area as the cuboid. To do this, he needs to form and solve an equation for p .



- (i) If the surface area of the cube is the same as the surface area of the cuboid, show that $2p^2 - 20p + a^2 = 0$.

Remember that the surface area of the cube is $6p^2$ cm².

- (ii) As mentioned in part (i), if the surface area of the cube is the same as the surface area of the cuboid, then $2p^2 - 20p + a^2 = 0$.

By using the discriminant (Δ), find the largest possible **whole** number value that a could take in this context.

Use this value of a to find the dimensions of both the cube and the cuboid.

Explain your reasoning clearly.

- (c) A calendar can be presented in the following way, where each day is given a number from 1 to 365. This is the beginning of a year's calendar:

M	T	W	TH	F	SA	SU
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31
32	33	34	35	36	37	38
39	40	41	42	43	44	45
46	47	48	49	50	51	52
53	54	55	56	57	58	59
60	61	62	63	64	65	66
67	68	69	70	71	72	73

Jo draws a 4-by-4 square on the calendar to check a claim that she heard:

“the sums of the diagonally opposite corners are always the same, no matter where you make your square”. In other words, when you add the numbers in the orange corners, it is the same as when you add the numbers in the blue corners.

Jo wonders if the claim will still be true no matter where she starts the square, so she begins an investigation using algebra:

A			

- (i) Use algebra to prove that, no matter where the 4-by-4 square is drawn on the calendar, **the sum of the orange corners must be the same as the sum of the blue corners.**

- (ii) Jo wonders about the **products** of the diagonally opposite corners: could they be the same?

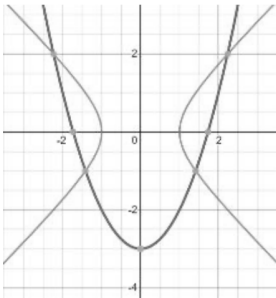
Use algebra to prove that it is **not** possible to draw any 4-by-4 square for which the products of the diagonally opposite corners are the same.

(c) Solve $\sqrt{2x+3} = 3x$.

(d) Suppose $Q(x) = fx^2 + gx + h$, where f , g , and h are constants. The “reciprocal polynomial” of $Q(x)$ is defined as $Q^*(x) = hx^2 + gx + f$, where the coefficients are in the reverse order.

(i) Find the solutions of the equation $Q(x) = Q^*(x)$.

- The roots of $Q^*(x) = 0$ are multiples of A and of B, i.e. the roots are kA and kB for some constant k .

(d)	Curve Two: $y = (x - 1)(x + 1) - 2$ So $y = x^2 - 3$ Curve One: $x^2 = y^2 + 1$, so substitute into Curve Two to give: $y = y^2 + 1 - 3 = y^2 - 2$ $y^2 - y - 2 = 0$ $(y - 2)(y + 1) = 0$ Either $y = 2$ so $x^2 = 5$ and $x = \pm\sqrt{5}$, so points of intersection are $(\pm\sqrt{5}, 2) = (\pm 2.24, 2)$ Or $y = -1$ so $x^2 = 2$ and $x = \pm\sqrt{2}$, so points of intersection are $(\pm\sqrt{2}, -1) = (\pm 1.41, -1)$. FYI: 	Correct expansion and simplification of Curve Two.	Forms one equation in one variable.	T1: Correct identification of two solutions only or of four x values. T2: Complete correct solution with correct mathematical statements.
-----	---	--	-------------------------------------	--

N0	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	Part Q correct	1 of u	2 of u	3 of u	1 of r	2 of r	T1	T2

Official assessment schedule extract · pages 2 and 3

Q THREE	Expected coverage	Achievement (u)	Merit (r)	Excellence (t)
(d)(i)	SA of cuboid $= 2[(p - a)(p + a) + 10(p - a) + 10(p + a)]$ $= 2[p^2 - a^2 + 20p]$ Thus $2p^2 - 2a^2 + 40p = 6p^2$ $0 = 4p^2 - 40p + 2a^2$ $0 = 2p^2 - 20p + a^2$	Finds expression for SA of cuboid and expands it (accept no use of factor of 2 for all sides).	Derives given equation clearly.	
(d)(ii)	For there to be solutions, $400 - 8a^2 > 0$ $50 > a^2$ Hence $0 < a < 7.1$. If $a = 7$, we solve $2c^2 - 20c + 49 = 0$ and obtain $p = 5.707$ or $p = 4.293$. However, one of the sides of the cuboid is $p - a$, meaning that it will be negative with either value of p obtained here. If $a = 6$, we solve $2p^2 - 20p + 36 = 0$ and obtain $p = 7.646$ or $p = 2.354$. The larger of these 2 values is a valid solution so the largest possible whole number value of a is 6. The cube has sides of 7.646 cm, and the cuboid is $1.646 \times 13.646 \times 10$ cm.	Evaluates discriminant.	Obtains inequality (or implies one) for a or a^2 .	T1: Obtains $a = 7$ and dimensions of 5.707 for the cube but does not go on from there. T2: Obtains correct dimensions for the cube and the cuboid.

N0	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	Part Q correct	1 of u	2 of u	3 of u	1 of r	2 of r	T1	T2

Official assessment schedule extract · pages 4 and 5

Q	Evidence	Achievement	Merit	Excellence
(c)(i)	Sum of orange corners: $A + A + 24 = 2A + 24$ [A + B] Sum of blue corners: $A + 21 + A + 3 = 2A + 24$ [(A+3) + (B - 3)] Therefore sum of orange corners = sum of blue corners, no matter where you start the square.	Correct algebraic evidence but no conclusion.	Two sums compared and conclusion explicitly drawn.	
(ii)	Product of orange corners: $A(A + 24) = A^2 + 24A$ Product of blue corners: $(A + 21)(A + 3) = A^2 + 24A + 63$ If these products are equal: $A^2 + 24A + 63 = A^2 + 24A$ ** So $63 = 0$ Which is impossible. Or a statement that 63 cannot equal zero. OR An argument based on the orange corners being A and B, and the blue corners being A + 3 and B - 3, leading to $3B - 3A - 9 = 0$ $B - A = 3$ ## This cannot true if B is on a different row, and, as this is not true, the products cannot be equal. [or equivalent arguments with different valid expressions for the corners]		Correct algebraic evidence up to line **. OR Simplified relationship between A and B (line ##) but no conclusion	Correct and complete algebraic reasoning. OR Correct algebraic evidence with conclusion.

Official assessment schedule extract · page 6

(d)	For a rectangle M wide and N tall: Sum of orange corners: $A + [A + (M - 1) + 7(N - 1)]$ $= 2A + M + 7N - 8$ Sum of blue corners: $[A + 7(N - 1)] + [A + (M - 1)]$ $= 2A + M + 7N - 8$ Both sums have the same expression so are always equal. Accept alternative approaches that are valid arguments.		Reasoning valid but M and / or N used for the corners. OR Correct algebraic evidence but no conclusion.	Correct and complete reasoning with conclusion stated.
-----	---	--	--	--

N0	N1	N2	A3	A4	M5	M6	E7	E8
----	----	----	----	----	----	----	----	----

Q	Evidence	Achievement	Merit	Excellence
(c)	$\sqrt{2x+3} = 3x$ $2x+3 = 9x^2$ $9x^2 - 2x - 3 = 0$ $x = 0.6991$ or $x = -0.4768$ (4sf)	Obtains correct quadratic.	Obtains both correct solutions.	
(d)(i)	$fx^2 + gx + h = hx^2 + gx + f$ $(f-h)x^2 + (h-f) = 0$ $(f-h)(x^2 - 1) = 0$ $x^2 = 1$ $x = 1$ or $x = -1$ Accept $\pm \sqrt{\frac{-(h-f)}{(f-h)}}$ or equivalent.		Correct working to obtain one solution only.	Both correct solutions obtained.

(ii)	Roots of $Q(x)$ are $x = \frac{-g \pm \sqrt{g^2 - 4fh}}{2f}$ Roots of $Q^*(x)$ are $x = \frac{-g \pm \sqrt{g^2 - 4hf}}{2h}$ If $A = \frac{-g - \sqrt{g^2 - 4fh}}{2f}$, then one of the roots of $Q^*(x)$ will be $kA = k \left(\frac{-g - \sqrt{g^2 - 4fh}}{2f} \right) = \frac{-g - \sqrt{g^2 - 4fh}}{2h}$ so $k = \frac{f}{h}$ OR If roots of $Q(x)$ are A and B, $AB = \frac{h}{f}$ If roots of $Q^*(x)$ are kA and kB, $(kA)(kB) = \frac{f}{h}$ So it follows that: $k^2 AB = k^2 \frac{h}{f} = \frac{f}{h}$ and $k^2 = \frac{f^2}{h^2}$ and $k = (\pm) \frac{f}{h}$ OR If roots of $Q(x)$ are A and B, $A+B = \frac{-g}{f}$ If roots of $Q^*(x)$ are kA and kB, $kA + kB = \frac{-g}{h}$ So it follows that: $k(A+B) = k \left(\frac{-g}{f} \right) = \frac{-g}{h} k$ and $k = \frac{f}{h}$	Correct expressions for all 4 roots obtained (may be combined) OR $AB = \frac{h}{f}$ OR $A+B = \frac{-g}{f}$	Correct expression involving k as function of f,g and / or h OR Finds $k = \frac{h}{f}$ [this results from saying the root of $Q(x)$ is kA and that of $Q^*(x)$ is A]	Finds $k = \frac{f}{h}$
------	--	---	---	---

NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	A valid attempt at one question.	1u	2u	3u	1r	2r	1t	2t